

Benefits of Marriage as a Search Strategy

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ABSTRACT

We propose and study an agent-based model of marriage. Agents hold private, asymmetric preferences over one another, drawn from a probability distribution, and search for better mates through a stochastic matching process, forming couples and breaking them up as they go. Marriage is modelled as a commitment device: an agent stops searching once the affinity with its partner exceeds a threshold Λ . We show that average utility in a society with marriage can exceed that of a society without it, and that the effect is maximised near $\Lambda \approx \sigma$, one standard deviation above the mean affinity. Part of our results can be summarised as what sounds like good advice: do not marry the first person you like and do not hold out for the love of your life, but do marry if you like your partner more than a sigma above average. Marriage also compresses the distribution of outcomes, so it raises equality as well as the mean. We formulate the model in the large-population limit, derive the stationary distribution of married couples in closed form, and delimit the regime in which the mean-field description is valid. To test the adequacy of the model for real societies we compare the evolution of the fraction of married agents with cohort data from England and Wales, and obtain good agreement.

KEYWORDS

Agent-based modelling; matching; search and stopping; social simulation; mean-field analysis

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1 INTRODUCTION

Consider the following situation: you are wondering whether to propose marriage to your partner. There are many aspects of marriage to consider, but you are mainly concerned with marriage as a *search strategy*. You like your partner, but you are not sure it is enough. After all, there may be someone else out there whom you would prefer even more, and you are unlikely to meet that person if you marry. Perhaps, from a search-strategy perspective, the best option is to avoid marriage and remain open to the chance of meeting a better match. However, if your relationship is symmetric in this regard, you may find yourself single because your partner has moved on. While marriage reduces some of the opportunities that would otherwise be available to you, it also protects you from being left. So you ask yourself: is getting married a good search strategy? Does the possibility of meeting someone with whom you

have a higher affinity outweigh the risk of being alone because your current partner has left?

A famous problem related to this subject is the stable matching problem [18]: the problem of finding a stable matching between two groups of agents with preferences over one another. Interpreting the groups as men and women, a matching is a bijection between the two, and it is stable if there is no man and woman who both prefer each other over their assigned mates. Since its inception there has been extensive research on the mathematical structure of stable matchings and on the associated algorithmic questions [18, 21, 25, 38]. The notion is fundamental to understanding a variety of markets, as surveyed in [35], and matching theory is a central tool in market design [1, 37].

When it comes to large societies, however, the concept of a stable matching may not be the right organising idea. In a society of millions it is all but certain that there exist two agents who would rather be together than with their respective mates. But that blocking pair may never meet. Whether they do, and what happens if they do, depends on the kind of relationship each is already in. If they are married, they are by assumption not open to the possibility. Otherwise, they may eventually match. To make these ideas precise, and to answer the question of the opening paragraph, we propose an agent-based model of marriage built on a random matching process and simple behavioural rules.

The model contains N agents with preferences over one another sampled from a fixed distribution. Initially all agents are single, and the system evolves for T discrete steps. At each step agents are paired at random. If two paired agents like each other more than their current partners—or more than being single—they form a new couple, and whoever is left behind becomes single again. Marriage is introduced as a stopping rule: once two agents like each other by more than an individual threshold Λ , they stop searching and are removed from the matching process for good.

We simulate the system for a range of Λ and track the average utility as a function of the number of steps. The answer to the question of the first paragraph can be summarised as what sounds like good advice: *do not marry the first person you like and do not search for the love of your life, but get married if you like your partner more than a sigma above average*. In addition, the model's output can be compared with data: the fraction of married agents at each step plays the role of the fraction of a birth cohort married by a given age, which is observable.

Our contributions are as follows.

- We define a minimal agent-based model of mate search in which marriage is a commitment device, and show that the average utility of a society with marriage can exceed that of a society without it, with an optimum near $\Lambda \approx \sigma$ (Section 5).
- We show that marriage simultaneously reduces the dispersion of outcomes, so the benefit is not purchased at the cost of equality—in the homogeneous model. Once agents differ in how much they are liked, the picture changes: the gain

accrues to the less desirable, and past a threshold the most desirable agents are made strictly worse off (Section 5.5).

- We formulate the dynamics in the large- N limit as a recursion on the utility density, recover it in closed form for the uniform model, and determine empirically where the mean-field description is exact and where it is not (Section 7).
- We derive the stationary distribution of married couples in closed form for an arbitrary affinity distribution, and give its mean (Section 7.5).
- We compare the model against centralised benchmarks—Gale–Shapley and the utilitarian optimum—and against cohort marriage data from England and Wales (Sections 6 and 8).

All results are reproducible; code and data accompany the paper.

2 RELATED WORK

The idea of using formal tools to study marriage is not new [6–8]. Following the taxonomy for models of the marriage market proposed in [12], our model fits best in the category of search models. For detailed presentations of search theory see [11, 34, 36, 41], and for search models of marriage in particular see [9, 10, 20, 31, 39, 40].

Our approach differs from these works in two respects. Instead of analysing matching properties for generic preferences, as most of the referenced work does, we assume a probability distribution over preferences and specify a concrete matching process. The motivation for adopting a distribution stems from our interest in the regime $N \gg T$, where the number of agents far exceeds the number of times any agent meets someone new. Compared with work that does not assume completely generic preferences [5, 28], we impose *less* structure on agents’ preferences in this way. Our matching process, in turn, is an attempt to model how people actually meet.

The model is therefore close in spirit to work in complexity economics [3, 4], and it illustrates a case in which one of conventional economics’ assumptions does not hold. The dynamics are driven by simple behavioural rules, but the resulting pattern of evolution is complex, and no agent has the computational power to work out which behaviour will lead to maximum average utility. Utility maximisation does not appear to be dictating the behaviour of agents in these scenarios.

There is also a literature on dynamic and multi-period notions of stability [2, 13–16, 19, 23, 26, 27, 29, 33], which addresses matching markets that evolve over time. Our results suggest that even these generalised notions may not be the right lens for the matching of couples in large societies, for a reason we make precise in Section 3.2: the system has no equilibrium to converge to. A different route to the large-population limit of a matching problem was taken in [17].

3 THE MATE-SEARCH MODEL

The mate-search model is the backbone of the marriage model of Section 4. It contains N agents $\mathcal{A} = \{1, \dots, N\}$ with preferences over one another described by a utility function [30]. This information is encoded in the *affinity matrix* $A \in \mathbb{R}^{N \times N}$, where A_{ij} is the utility for agent i of being in a couple with agent j . Note that A is not symmetric in general, because the utility for agent i of

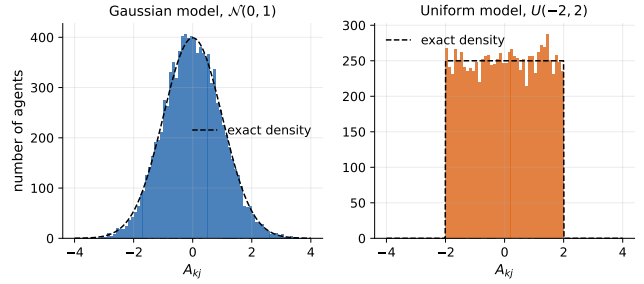


Figure 1: Histogram of the values $\{A_{kj} : j \neq k\}$ for a single agent k , with bin size 0.1, in the Gaussian and uniform models with $N = 10,000$, $\mu = 0$ and $\sigma = 1$. Dashed: the generating density.

being with agent j need not be the same as the reverse; and A_{ij} is not positive in general, as it is common to meet people one would rather be single than date. We refer to the collection of agents as the *society*.

3.1 Affinities

In large societies the distribution that generates the affinity matrix matters more than its actual values. The preferences of agent k concerning the rest of society are encoded by $\{A_{kj} : j \neq k\}$, whose histogram (Figure 1) follows some probability distribution. As with many human traits, it may be reasonable to assume these values are normally distributed, $A_{kj} \sim \mathcal{N}(\mu_k, \sigma_k)$.

One might think it more realistic to let different agents have distributions with different parameters. Note, however, that the k -th *column*, $\{A_{jk} : j \neq k\}$, encodes how agent k is liked by society. If agents had preferences given by Gaussians with different parameters, the distribution generating the columns would not in general be Gaussian. To keep rows and columns consistently Gaussian we therefore assume that all off-diagonal entries are generated by the same distribution, $A_{kj} \sim \mathcal{N}(\mu, \sigma)$ for all $j \neq k$. We call this the *Gaussian model*. For comparison, and to make the analytical work of Section 7 concrete, we also study the *uniform model*, $A_{ij} \sim U(\mu - 2\sigma, \mu + 2\sigma)$. We return to genuinely heterogeneous societies in Section 5.5.

The diagonal elements of the affinity matrix are the utilities of being single. We also take these to be generated by a distribution, with density p_0 . The mean of the diagonal is the average utility when all agents are single; the mean of the off-diagonal entries is the average utility when agents form random couples. The relationship between the two therefore encodes how being single compares with dating a random member of society. We do not believe these two quantities are generically equal, but for concreteness we present our main results assuming they are, and show in Section 5.4 that the conclusions survive when they are not. In the analytical treatment of Section 7 the two distributions are kept separate throughout: we write p_0 for the density generating the diagonal and p_A for the density generating the off-diagonal entries.

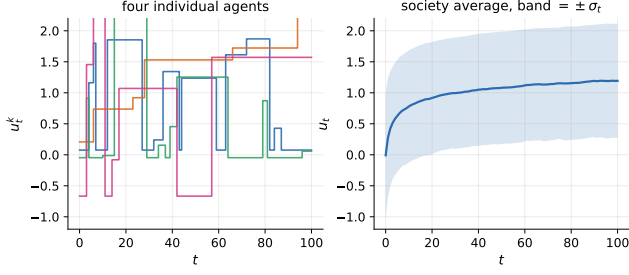


Figure 2: Left: utility of four agents, distinguished by colour. Right: average utility with a band of one standard deviation. Gaussian model $\mathcal{N}(0, 1)$, $N = 10,000$, no marriage. The uniform model $\mathcal{U}(-2, 2)$, which has the same variance, is visually indistinguishable.

3.2 Dynamics

The dynamics happen in discrete steps $t \in \mathbb{N}$. At each step all agents are paired at random: a random perfect matching, that is, an involution without fixed points

$$\pi_t : \{1, 2, \dots, N\} \rightarrow \{1, 2, \dots, N\}, \quad (1)$$

drawn uniformly. The agent k meets at step t is $\pi_t(k)$. We write u_t^k for the utility of agent k at step t .

Suppose first that agent k is single at t , so $u_t^k = A_{kk}$, and meets agent $l = \pi_{t+1}(k)$. The two form a couple if they like each other enough:

$$u_{t+1}^k = \begin{cases} A_{kl}, & \text{if } A_{kl} > u_t^k \text{ and } A_{lk} > u_t^l, \\ u_t^k, & \text{otherwise.} \end{cases} \quad (2)$$

Now suppose agent k is in a couple with agent m at t , so $u_t^k = A_{km}$ and $u_t^m = A_{mk}$. At step $t + 1$ let k meet $l = \pi_{t+1}(k)$ and m meet $n = \pi_{t+1}(m)$. If the affinity between newly met agents exceeds that between the mates, a new couple forms and the old one breaks apart:

$$u_{t+1}^k = \begin{cases} A_{kl}, & \text{if } A_{kl} > u_t^k \text{ and } A_{lk} > u_t^l, \\ A_{km}, & \text{if } A_{mn} > u_t^m \text{ and } A_{nm} > u_t^n, \\ u_t^k, & \text{otherwise.} \end{cases} \quad (3)$$

Both comparisons are made against the utilities as of the start of the step. We start the system with all agents single, $u_0^k = A_{kk}$. A couple here is not a married couple; marriage is introduced in Section 4.

Running the simulation for T steps produces an $N \times T$ array $\{u_t^k\}$. As we are interested in large societies, $T \ll N$, we focus on the average utility and its dispersion,

$$u_t \equiv \frac{1}{N} \sum_{k=1}^N u_t^k, \quad \sigma_t \equiv \sqrt{\frac{1}{N} \sum_{k=1}^N (u_t^k)^2 - \left(\frac{1}{N} \sum_{k=1}^N u_t^k\right)^2}. \quad (4)$$

All moments carry aggregate information; Section 7 treats the higher ones. The difference between u_t^k and u_t is visible in Figure 2: individual trajectories jump up when an agent finds someone better and crash back to A_{kk} when an agent is left, while the average rises smoothly.

3.3 Scale Invariance

The results we present were obtained with $\mu = 0$ and $\sigma = 1$. When all entries of the affinity matrix come from the same distribution there is a symmetry that carries them to generic μ and σ . Equations (2) and (3) depend only on inequalities between entries of A , so the dynamics are unchanged under the affine map $A_{ij} \mapsto \sigma A_{ij} + \mu$. The utility of each agent transforms the same way:

$$\text{if } (\mu = 0, \sigma = 1) \mapsto u_t^k \text{ then } (\mu, \sigma) \mapsto \sigma u_t^k + \mu. \quad (5)$$

It therefore suffices to read 0 as μ and to interpret the vertical axis of every figure in units of σ .

3.4 No Equilibrium to Converge To

In large societies the number of new people we meet over a lifetime is much smaller than the size of society, $T \ll N$. The asymptotic limit $T \rightarrow \infty$ is correspondingly not the object of interest. This resonates with Keynes’s remark that “in the long run we are all dead” [24], read as the advice that asymptotic limits and equilibrium states are not always relevant.

More than that: in the model without marriage, the limit $T \rightarrow \infty$ does not approach any equilibrium at all. Consider three agents A , B and C , where A has highest affinity with C , B with C , and C with A . At each step one of them is left out, and they remain in an eternal dance—a situation not unknown in modern relationships. Because the system never settles, even generalised notions of stability [14, 16, 23, 27] may not be the right tool for understanding how couples match in large societies.

4 THE MARRIAGE MODEL

The marriage model extends the mate-search model as follows. We introduce the *proposal vector* $\vec{\Lambda} \in \mathbb{R}^N$, where Λ_k encodes how much agent k needs to like another agent before proposing marriage. If the affinity within a couple exceeds both members’ thresholds, the couple stops meeting new agents. We then say the couple is *married*:

$$\text{the couple } (k, l) \text{ is married at } t \iff A_{kl} > \Lambda_k \text{ and } A_{lk} > \Lambda_l. \quad (6)$$

Let $\mathcal{M}_t \subseteq \{1, \dots, N\}$ be the set of agents married at time t . The matching at time $t + 1$ does not include them:

$$\pi_{t+1}^\Lambda : \{1, \dots, N\} \setminus \mathcal{M}_t \rightarrow \{1, \dots, N\} \setminus \mathcal{M}_t. \quad (7)$$

Because married agents are never matched again, they can never be poached, and a married couple never breaks. Consequently $\mathcal{M}_t$ is non-decreasing in t , and $\Lambda = \infty$ recovers the mate-search model exactly.

As before, what matters for a large society is the distribution generating $\vec{\Lambda}$, and a reasonable assumption is $\Lambda_k \sim \mathcal{N}(\Lambda, \sigma_\Lambda)$. The values Λ and σ_Λ are exogenous: they characterise the behaviour of the agents of a given society. Different societies—or the same society at different times—will have different values of Λ , a point we return to in Section 8. To keep the investigation tractable we take the limit $\sigma_\Lambda \rightarrow 0$ throughout, so $\Lambda_k = \Lambda$ for all k .

The scaling relation (5) continues to hold provided Λ is shifted and rescaled along with the affinities. If u_t^k is obtained for $\mu = 0$,

Table 1: Average utility u_{100} at the end of the run, by threshold. Mean $\pm$ standard error over eight societies of $N = 10,000$.

Threshold	Gaussian $\mathcal{N}(0, 1)$	Uniform $U(-2, 2)$
$\Lambda = \infty$ (no marriage)	1.191 ± 0.004	1.269 ± 0.003
$\Lambda = 1.5$	1.298 ± 0.003	1.519 ± 0.001
$\Lambda = 1$	1.447 ± 0.003	1.543 ± 0.001
$\Lambda = 0.5$	1.285 ± 0.002	1.391 ± 0.001
$\Lambda = 0$	1.075 ± 0.003	1.255 ± 0.002
$\Lambda = -2$	0.868 ± 0.002	0.988 ± 0.002

$\sigma = 1$ and threshold Λ , then

$$(\mu, \sigma, \sigma\Lambda + \mu) \mapsto \sigma u_t^k + \mu. \quad (8)$$

So the results below are generic in μ and σ , with Λ measured in units of σ above the mean.

5 RESULTS

Unless stated otherwise, all simulations use $N = 10,000$ agents, $T = 100$ steps, $\mu = 0$ and $\sigma = 1$. Where error bars are quoted they are standard errors over eight independent societies. When we compare a society with marriage against one without, both are run on the *same* affinity matrix with independent matchings, so that the difference between the two is not an artefact of sampling A .

5.1 Marriage Can Raise Average Utility

Figure 3 and Table 1 compare the average utility for different values of Λ . The most significant lessons of the model can be read off from them.

Two limits bracket the behaviour. If $\Lambda > A_{ij}$ for all i, j , marriage never triggers and the dynamics are those of the mate-search model; for sufficiently large Λ at finite N , the two models coincide. If instead $\Lambda < A_{ij}$ for all i, j , every agent marries the first agent it likes, and the average utility drops well below the no-marriage baseline—at $\Lambda = -2$, to 0.868 against 1.191. The lesson is that one should not marry the first person one likes.

Between the two extremes something more interesting happens. For Λ near σ the average utility is consistently *larger* than in the society without marriage: 1.447 against 1.191 in the Gaussian model, a gain of 0.26σ , and 1.543 against 1.269 in the uniform model. It is therefore reasonable to conclude that one should marry if one likes one’s partner more than a sigma above average. Note also that the curves cross: the value Λ_t maximising average utility depends on the horizon t , and a society that expects a long search should be more demanding than one that does not.

To clarify the interpretation, consider the following scenario. During our lives we can take every opportunity to meet others, pursuing the dream of finding the greatest love of our lives. Alternatively, having found someone who makes us happy by more than a certain amount, we can stop searching. The model says that on average it is better to stick with someone good enough than to keep looking. Together with the $\Lambda = -2$ result, this justifies the advice advertised above: do not marry the first person you like and do not search for the love of your life, but get married if you like your partner more than a sigma above average.

There is a second interpretation that motivated this model. Suppose you are dating and must decide whether to engage in monogamy or non-monogamy. The most salient difference is that in non-monogamy you and your mate remain open to meeting others. Many features of your preferences bear on the decision: how much you enjoy meeting new people, whether you have time for both a mate and sporadic matches, how jealous you feel. In addition, the openness characteristic of non-monogamous relationships increases the chance of meeting someone with whom either you or your mate has a higher affinity. You may then wonder whether the odds of finding someone better outweigh the risk of being left. If one reads our couples as non-monogamy and our married couples as monogamy, the model addresses this question and gives a negative answer. If you are in all other respects unsure which you prefer, the benefit of marriage as a search strategy disclosed here should tip the balance towards monogamy. This interpretation may grow in relevance as the number of people who experience non-monogamous relationships increases [22]. Throughout the rest of the paper we stick to the first interpretation.

5.2 Marriage Also Reduces Inequality

It is instructive to plot the evolution of u_t together with its dispersion, as in Figure 4. The society with marriage at $\Lambda = \sigma$ has both a higher mean and a *lower* variance: in the Gaussian model the standard deviation at $t = 100$ falls from 0.914 to 0.645, and in the uniform model from 0.838 to 0.371. If agents value equality with respect to relationships, this is a second benefit of marriage on top of the first. The mechanism is visible in the full distributions (Figure 5): marriage imposes a floor, because every married agent is above Λ by construction, and it drains the mass of long-term singles sitting at their A_{kk} .

5.3 The Price: A Shorter Search

The benefit is bought with something. Figure 6 counts the distinct partners each agent has over the run. Without marriage an agent cycles through 5.15 partners on average in the Gaussian model; with marriage at $\Lambda = 1$, 4.04. In the uniform model the reduction is sharper, from 5.00 to 2.76. An agent who does not marry ends up on average with someone it likes less, but it may nevertheless derive a surplus from having participated in more relationships—a surplus this model does not represent. Conversely, it is in general painful to break up, and that loss is not represented either, though accounting for it would add a further benefit to the society with marriage. Overall, the evidence suggests that marriage is beneficial in a number of respects [42].

5.4 Robustness: Being Single Need Not Be the Same Draw

In Section 3.1 we assumed for concreteness that the diagonal and off-diagonal elements of the affinity matrix come from the same distribution. One might reasonably worry that the results depend on it. To address this we simulate the model with $A_{kk} = 0$ for all k —being single is worth exactly nothing—and $A_{ij} \sim \mathcal{N}(\mu, 1)$ for $i \neq j$, sweeping μ , with the threshold tracking the distribution at $\Lambda = \mu + \sigma$. Negative μ means most people are worse than solitude; positive μ means almost anyone beats it. Note that $A_{kk} = 0$ is

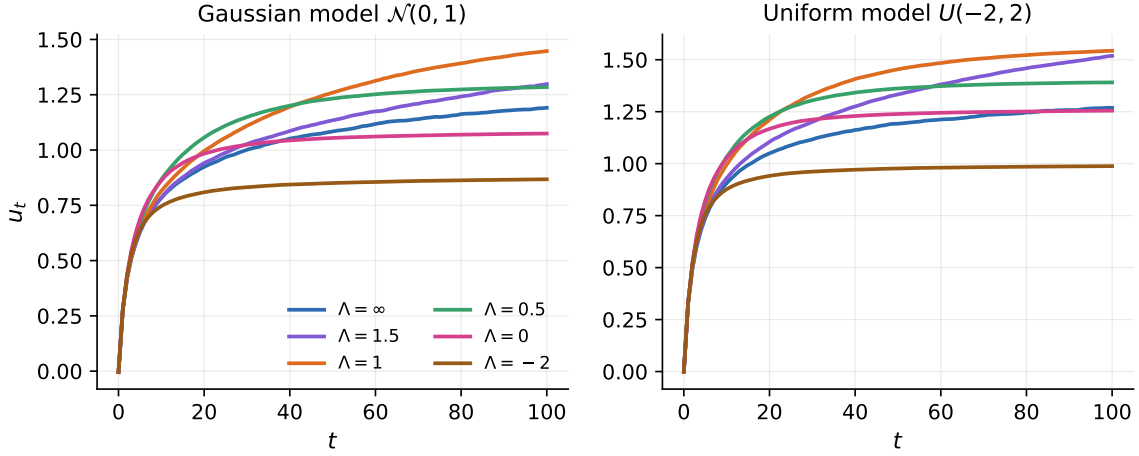


Figure 3: Average utility evolution in the Gaussian and uniform models with $N = 10,000$ for different values of Λ ; $\Lambda = \infty$ denotes the society without marriage. Mean over eight independent societies, with standard-error bands.

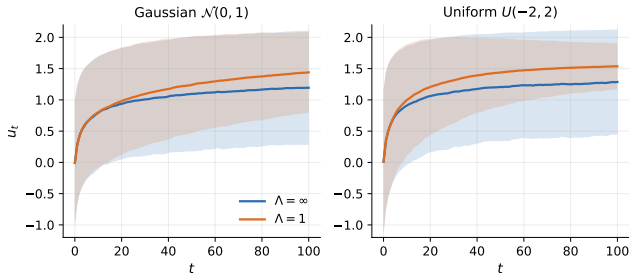


Figure 4: Average utility with marriage at $\Lambda = 1$ (orange) and without marriage (blue), with bands of one standard deviation, for $N = 10,000$. Both societies share the same affinity matrix.

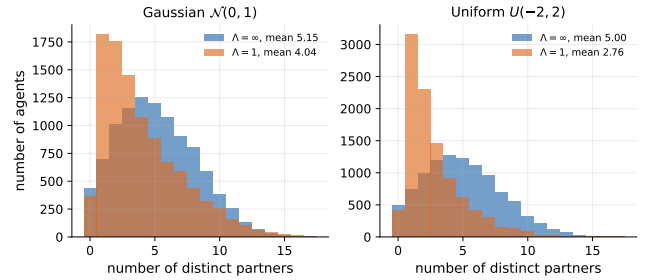


Figure 6: Distinct partners per agent over 100 steps, with and without marriage, $N = 10,000$.

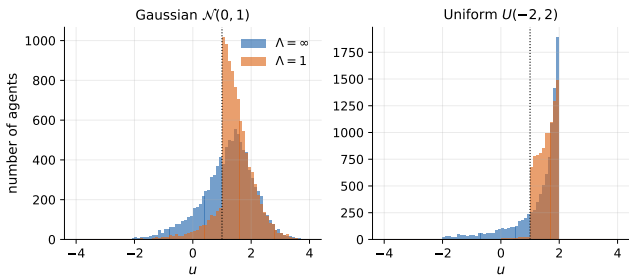


Figure 5: Histogram of $\{u_{100}^k\}$ with bin size 0.1 for the society with marriage at $\Lambda = 1$ (orange) and without marriage (blue). Dotted line: the threshold $\Lambda = 1$.

obtained formally by taking $\sigma_s \rightarrow 0$ in the distribution generating the diagonal, and that in this limit there is no loss of generality in setting $\sigma = 1$ off the diagonal, since (2) and (3) remain invariant under rescaling of A_{ij} alone.

Figure 7 shows the outcome. Writing $f(\mu) \equiv u_M(\mu) - u(\mu)$ for the gap between the society with marriage and the society without, we find $f(\mu) > 0$ throughout, and f non-decreasing in μ : $f(-2) = +0.001$, $f(-1) = +0.027$, $f(0) = +0.283$, $f(1) = +0.565$, $f(2) = +0.739$. The two ends of the sweep also tell us what happens outside it. For $\mu < -2\sigma$ an agent is unlikely to meet anyone it prefers to being single, and the system behaves much as at $\mu = -2$. For $\mu > 2\sigma$ agents prefer almost any partner to solitude, as at $\mu = 2$, and the dynamics are similar up to the change of scale set by μ . Since the gap is positive throughout, the benefit of marriage is a robust feature of the model, and the advice extracted from it holds in more general contexts.

5.5 Who Gains, and Who Loses

A further objection is that the assumption of a homogeneous society is plainly false. What is interesting is that at finite N homogeneity is already broken: each agent's column $\{A_{jk} : j \neq k\}$ is a different collection of numbers, even though all are drawn from one distribution. One can select the 1% of agents whose column mean is largest—the most liked agents of the society—and, conversely, the 1% least liked, and follow each group separately.

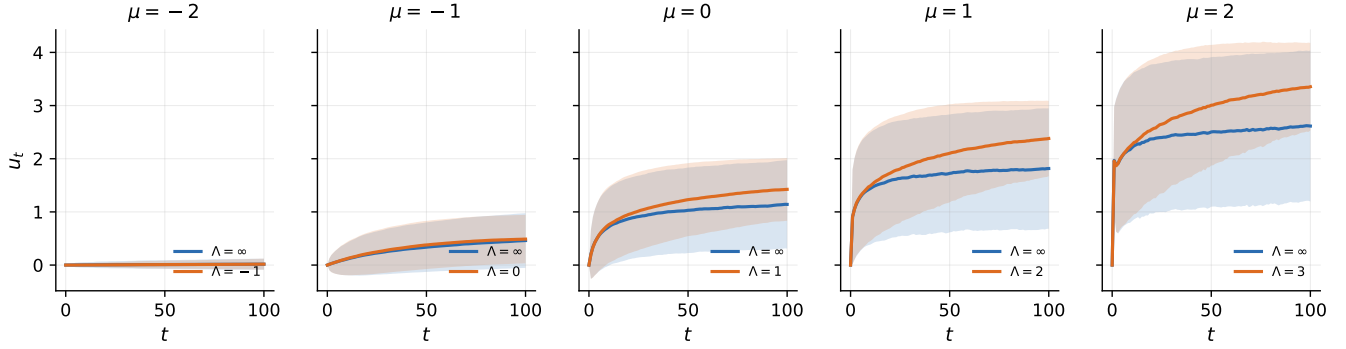


Figure 7: Average utility with $A_{kk} = 0$ for all k and $A_{ij} \sim \mathcal{N}(\mu, 1)$ for $i \neq j$, $N = 10,000$. Orange: with marriage at $\Lambda = \mu + 1$. Blue: without marriage.

Table 2: Marriage premium $u_M - u$ at $t = 100$ by desirability spread σ_q , for the most and least liked 10% of agents. Total variance of A_{ij} held at 1. Mean $\pm$ standard error over six societies.

σ_q	Least liked 10%	Most liked 10%
0.0	$+0.250 \pm 0.015$	$+0.281 \pm 0.014$
0.2	$+0.249 \pm 0.014$	$+0.108 \pm 0.013$
0.4	$+0.169 \pm 0.009$	-0.051 ± 0.013
0.6	$+0.075 \pm 0.008$	-0.184 ± 0.009
0.8	-0.000 ± 0.003	-0.251 ± 0.008

This, however, is asking more of finite-size effects than they can deliver. With $N = 10,000$ i.i.d. entries, column means are spread by only $1/\sqrt{N} \approx 0.01\sigma$, so the “1% most liked” are more liked by about 0.03σ . Over six independent societies the marriage premium is $+0.219 \pm 0.045$ for the least liked and $+0.254 \pm 0.043$ for the most liked: indistinguishable. In the homogeneous model there is no such effect to find.

To ask the question properly the model has to be given genuine *vertical* heterogeneity. We therefore add a per-agent desirability q_k to every entry of agent k ’s column,

$$A_{ij} = \varepsilon_{ij} + q_j, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon), \quad q_j \sim \mathcal{N}(0, \sigma_q), \quad (9)$$

holding the total variance $\sigma_\varepsilon^2 + \sigma_q^2 = 1$ fixed so that runs at different σ_q remain comparable. Now some agents really are more widely liked than others, and σ_q tunes how much.

Figure 8 and Table 2 show the result. The paper’s original intuition is vindicated: marriage does help the less desirable more. But the effect is stronger than “more”. Past $\sigma_q \approx 0.4$ the premium for the most desirable agents turns *negative*: they are made strictly worse off by living in a society that marries. The reading is straightforward once stated. Marriage in this model is insurance against being left. Agents whom everyone likes are rarely at risk of being left, so the insurance is worth little to them, while the option value they give up—the chance of meeting someone even better, which for them is a realistic prospect—is worth a great deal.

This matters for how the advice of Section 5 should be read. The benefit of marriage is not society-wide once agents differ in desirability; there is a distributional conflict, and the agents who do best out of the search process are precisely the ones with an incentive to defect from the convention. A society-level threshold Λ is therefore not obviously self-enforcing, which is a question this model is well placed to ask and which we leave for future work.

5.6 Men, Women, and Other Partitions

A possible point of concern is that we did not distinguish men from women, as in the original marriage problem [18]; one might think we are dealing with the stable roommates problem instead. The distinction turns out not to matter here. We can partition the society in two and match only across the partition. The evolution of average utility for such a system is indistinguishable from Figure 4: averaging over four societies, the two-sided variant differs from the one-sided one by less than 0.03, well inside the run-to-run spread.

One can even partition the society into groups of different sizes, and the results are still very similar. In that case the matching in (1) is a random injection from the smaller partition into the larger, and a random subset of the larger group sits out each step. Several further variations are possible: one could select a subset of one partition and allow it to match within itself as well, which admits a reading as bisexual agents. This flexibility is in line with the diversity of modern relationships, and contrasts with results for the matching problem—many of them, including Gale–Shapley, depend on the two sides having the same number of elements [38].

6 COMPARISON WITH CENTRALISED BENCHMARKS

It is instructive to compare the average utility attained by the dynamics with what a central planner could achieve on the same affinity matrix. We use three reference points, all computed by splitting the society into two equal halves and matching across them, which is the setting Gale–Shapley is defined for: random pairing, which is the floor; the proposer-optimal stable matching [18], with

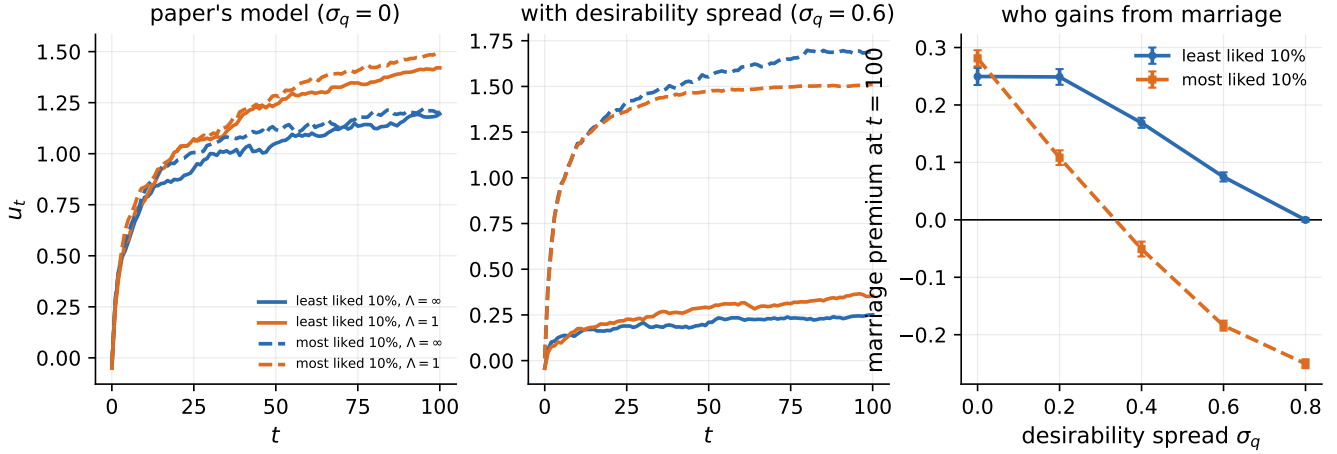


Figure 8: Marriage premium at $t = 100$ for the most and least liked 10% of agents. Left: the homogeneous model, $\sigma_q = 0$, where the two groups are indistinguishable. Centre: with desirability spread $\sigma_q = 0.6$. Right: the premium as a function of σ_q , mean $\pm$ standard error over six societies. Past $\sigma_q \approx 0.4$ the premium for the most desirable agents is negative.

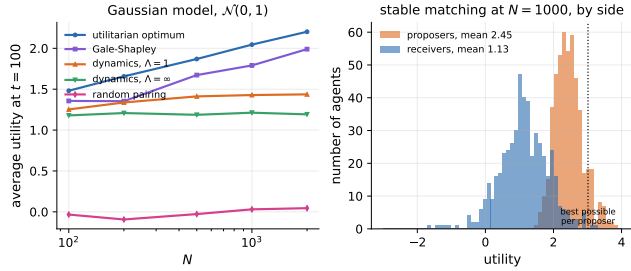


Figure 9: Left: average utility at $t = 100$ for the dynamics, against random pairing, the stable matching and the utilitarian optimum, as a function of N . Right: the two sides of the stable matching at $N = 1000$ do very different things.

average utility u_{G-S} ; and the matching π^* maximising total surplus,

$$u(\pi) = \frac{1}{2} \sum_{i=1}^N (A_{i\pi(i)} + A_{\pi(i)i}), \quad (10)$$

which is Becker’s paradigm of transferable utility [6, 7]. Such a π^* need not be stable, but by construction it maximises average utility.

A remark on computing π^* . Naively there are $N!$ matchings to consider, which for $N \approx 1000$ is a number with 2568 digits. But maximising (10) is a linear assignment problem on the symmetrised surplus $A_{ij} + A_{ji}$, so the Hungarian algorithm solves it exactly in $O(N^3)$ time. We report it up to $N = 2000$.

Figure 9 collects the comparison. At $N = 1000$ in the Gaussian model the dynamics reach 1.213 without marriage and 1.429 with marriage at $\Lambda = 1$; the stable matching reaches $u_{G-S} = 1.791$ and the utilitarian optimum 2.045. Random pairing gives 0.030, i.e. μ , as it must. The planner’s advantage grows with N , since both centralised benchmarks have more to work with as the society grows while the dynamics are limited by T .

Two observations follow. First, a stable matching does much better than the decentralised process, which is one indication that the concept of stability is not what organises matching in large societies. Second, and more usefully, the fact that u_{G-S} exceeds u_t at $t \approx 100$ implies that the market for relationships in large societies is substantially sub-optimal. If affinities could be inferred from manifest, correlated properties—age, personality, political leaning, level of education—then a central mechanism could propose a stable matching and society’s average utility with respect to relationships would be considerably higher.

The right panel of Figure 9 adds a caveat that bears on how large that gap really is. A proposer-optimal stable matching is far from symmetric between the two sides: at $N = 1000$ the proposing side averages 2.454 while the receiving side averages 1.128, and u_{G-S} is the average of the two. For reference, the mean best-possible partner for a proposer is 3.015, so no matching of any kind can place the two-sided average near it. Any claim about the size of the planner’s advantage should therefore say which side of the market it is talking about.

7 THE LARGE-POPULATION LIMIT

So far we have focused on the average utility and its variance, which are aggregates of the set $\{u_t^k\}$. That set carries a great deal more information, as the histograms of Figure 5 show. A convenient way to collect it is to introduce a random variable U_t taking the values $\{u_t^k\}$ each with probability $1/N$. The average utility is then $\mathbb{E}[U_t]$ and the variance $\mathbb{E}[U_t^2] - \mathbb{E}[U_t]^2$, where

$$\mathbb{E}[U_t^n] \equiv \frac{1}{N} \sum_{k=1}^N (u_t^k)^n. \quad (11)$$

This makes precise what is lost by looking only at the mean and variance: the higher moments of U_t .

At $t = 0$ we have $u_0^k = A_{kk} \sim U_0$, with known density p_0 , so in the limit $N \rightarrow \infty$,

$$\mathbb{E}[U_0^n] = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N A_{kk}^n = \int_{-\infty}^{\infty} u^n p_0(u) du, \quad (12)$$

and we have complete information about the system. The density $p_t(u)$ that generates the values $\{u_t^k\}$ is thus more fundamental than the values themselves. Given p_t one obtains $\mathbb{E}[U_t^n]$ by integration; given all the moments one recovers p_t by computing the characteristic function

$$\mathbb{E}[e^{ikU_t}] = \int_{-\infty}^{\infty} e^{iku} p_t(u) du = \sum_{n=0}^{\infty} \mathbb{E}[U_t^n] \frac{(ik)^n}{n!} \quad (13)$$

and inverting the Fourier transform. Finding p_{t+1} from p_t is therefore equivalent to finding $\mathbb{E}[U_{t+1}^n]$ from $\mathbb{E}[U_t^n]$, and that is the task of this section.

Before starting, one caveat. Although more tractable and in a sense more fundamental, the continuum version lacks a feature that the finite implementation shares with reality. At $N \rightarrow \infty$ the society is genuinely homogeneous, whereas at finite N some agents are more liked than others—and, as Section 5.5 showed, in real societies they are more liked by considerably more than sampling noise would give. The analysis of Section 5.5 has no counterpart in the continuum theory as set up here.

7.1 Decomposition into Couples and Singles

Agents are either coupled or single at each step, and the two behave differently, so p_t decomposes as

$$p_t(u) = b_t p_{ct}(u) + r_t p_{st}(u), \quad (14)$$

with p_{ct} and p_{st} the utility densities of coupled and single agents and b_t, r_t their population shares, $b_t + r_t = 1$. Introducing random variables C_t and S_t for couples and singles, (14) is equivalent to

$$\mathbb{E}[U_t^n] = b_t \mathbb{E}[C_t^n] + r_t \mathbb{E}[S_t^n]. \quad (15)$$

At $t = 0$ all agents are single, so $b_0 = 0, r_0 = 1$ and $S_0 \sim U_0$.

Write F_A for the cumulative distribution function of p_A and $\bar{G}_A = 1 - F_A$ for its survival function, so that $\bar{G}_A(x)$ is the probability that a random agent's affinity for you exceeds x . Three scalars recur throughout:

$$\alpha_t \equiv \mathbb{E}_{L \sim U_t} [\bar{G}_A(L)], \quad (16)$$

$$\gamma_t \equiv \mathbb{E}_{M \sim C_t} [\bar{G}_A(M)], \quad (17)$$

$$\beta_t \equiv 1 - \alpha_t \gamma_t. \quad (18)$$

In words: α_t is the probability that a random agent accepts you, γ_t the same restricted to agents currently in a couple, and β_t the probability that your own partner is *not* poached this step.

7.2 Warm-Up: The First Step

At $t = 1$ all agents start single, so only α_0 is needed. Two single agents K and L form a couple if each likes the other more than being single, an event of probability

$$c_0(K, L) \equiv \bar{G}_A(K) \bar{G}_A(L), \quad (19)$$

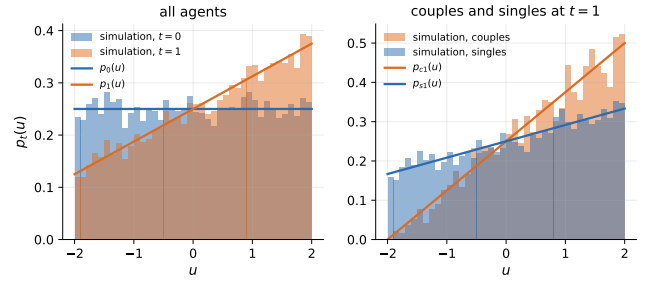


Figure 10: Exact densities after one step in the uniform model, against simulated histograms at $N = 10,000$. Left: all agents at $t = 0$ and $t = 1$. Right: couples and singles separately at $t = 1$.

and the fraction of agents in a couple at $t = 1$ is $b_1 = \mathbb{E}[c_0(K, L)]$ with $K, L \sim U_0$. For the homogeneous models, in which $p_0 = p_A$, this gives

$$b_1 = \frac{1}{4}, \quad r_1 = \frac{3}{4}. \quad (20)$$

In a typical simulation of the uniform model with $N = 10,000$ we obtain $b_1^{\text{exp}} = 0.2450$, and at $N = 200,000$, $b_1^{\text{exp}} = 0.2502$.

Conditioning on the couple forming, the new utility of K is distributed as p_A restricted to lie above K 's previous utility. Taking the expectation over $K \sim S_t$ and $L \sim U_t$ gives the densities directly:

$$b_1 p_{c1}(u) = \alpha_0 p_A(u) F_{S_0}(u), \quad (21)$$

$$r_1 p_{s1}(u) = p_{s0}(u) [\alpha_0 F_A(u) + 1 - \alpha_0]. \quad (22)$$

The second line says a single stays single either because the agent it met rejected it, or because it rejected the agent it met. For the uniform model, where $p_0 = p_A = 1/(4\sigma)$ on $[-2\sigma, 2\sigma]$, this evaluates to

$$p_{c1}(u) = \frac{2\sigma + u}{2\sigma} p_0(u), \quad p_{s1}(u) = \frac{6\sigma + u}{6\sigma} p_0(u), \quad (23)$$

whose weighted sum is

$$p_1(u) = b_1 p_{c1}(u) + r_1 p_{s1}(u) = \frac{4\sigma + u}{4\sigma} p_0(u). \quad (24)$$

Figure 10 compares these against the simulated histograms.

The mean after one step follows in closed form for both models:

$$\mathbb{E}[U_1] = \begin{cases} \frac{\sigma}{2\sqrt{\pi}}, & \text{Gaussian,} \\ \frac{\sigma}{3}, & \text{uniform,} \end{cases} \quad (25)$$

both consistent with the scaling relation (5). The simulation at $N = 10,000$ gives 0.2785 against $1/(2\sqrt{\pi}) = 0.2821$.

7.3 Evolution Equations

For $t \geq 1$ some agents are coupled, and a coupled agent has two ways to end the step still coupled: it forms a new couple, or it keeps the one it has, which requires that its partner was not poached.

Collecting the cases and writing the result directly in density form,

$$b_{t+1} p_{c,t+1}(u) = \alpha_t p_A(u) [b_t F_{C_t}(u) + r_t F_{S_t}(u)] + \beta_t b_t p_{ct}(u) [\alpha_t F_A(u) + 1 - \alpha_t], \quad (26)$$

$$r_{t+1} p_{s,t+1}(u) = \alpha_t \gamma_t \beta_t b_t p_0(u) + r_t p_{st}(u) [\alpha_t F_A(u) + 1 - \alpha_t]. \quad (27)$$

The first line of (26) covers new couples, formed by an agent—coupled or single—out-bidding whatever it had, with the new utility drawn from p_A above the old one. The second line covers survival. The first line of (27) is the inflow of agents whose partner left them, who revert to their single utility and so re-enter distributed as p_0 ; the second is singles who stayed single. Integrating recovers b_{t+1} and r_{t+1} , and since $c_0 + s_0 = 1$ identically, $b_{t+1} + r_{t+1} = b_t + r_t$, so probability is preserved by induction.

For the uniform model every density stays a polynomial, and the recursion can be carried out exactly in rational arithmetic. At $t = 2$,

$$b_2 = \frac{1861}{5184}, \quad r_2 = \frac{3323}{5184}, \quad (28)$$

with

$$p_{c2}(u) = \frac{3(2\sigma + u)(2258\sigma + 335u)}{7444(2\sigma)^3}, \quad (29)$$

$$p_{s2}(u) = \frac{3233(2\sigma)^2 + 3672\sigma u + 270u^2}{6646(2\sigma)^3}. \quad (30)$$

Further steps can be computed, but the coefficients grow explosively: the exact b_4 is a ratio of 35-digit integers. Figure 11 plots p_t for $t = 0, \dots, 4$ and compares p_2 with the simulated histogram.

7.4 Where the Mean-Field Description Is Valid

Equations (26) and (27) were derived by an intuitive argument, and it is natural to ask whether they really describe the $N \rightarrow \infty$ limit. Because the affinity matrix need not be stored explicitly—entries can be generated on demand from a counter-based hash of (i, j) , which keeps them fixed and reproducible at $O(N)$ memory—we can simulate societies of $N = 10^6$, that is, an affinity matrix with 10^{12} entries. At that size sampling noise is negligible and any residual disagreement belongs to the theory.

The answer, shown in Figure 12, is mixed. The occupancy is predicted correctly: b_2 converges to $1861/5184$ as $N^{-1/2}$, and at $N = 10^6$ agrees to 0.12%. The first step is predicted correctly in every respect, with u_1 accurate to 0.13%. But from $t = 2$ onwards the mean utility is systematically over-predicted—by 3.6% at $t = 2$, 5.3% at $t = 3$, and around 8% by $t = 100$ —and the discrepancy does *not* shrink as N grows from 10^3 to 10^6 . It is a bias, not a finite-size effect.

The source is an independence assumption. In deriving (26) the two members of a couple are treated as independent draws from C_t . They are not: a couple exists precisely because each member cleared the other's threshold, so their utilities are correlated from the moment it forms, and the correlation compounds as the couple survives successive steps. Population shares are insensitive to this over a single step, which is why b_2 comes out right; the shape of the density is not. A correct treatment would track the joint density of a couple, $p_{ct}(u, v)$, rather than its marginal—at the cost of turning a one-dimensional recursion into a two-dimensional one. We leave

this to future work, and read Figure 12 as delimiting where the present equations may be trusted: exactly at $t = 1$, for occupancy at $t = 2$, and as an upper bound on u_t thereafter.

7.5 The Distribution of Married Couples

In the model with marriage, married couples decouple from the system: they are never selected by the matching (7), so they never break, and once an agent is married its utility is frozen. This alone determines the utility distribution of married couples, and shows it to be the same at every step. As the system evolves an ever larger fraction of agents is married, until in the limit all are, so the asymptotic distribution

$$p_\infty(u) \equiv \lim_{t \rightarrow \infty} p_t(u) \quad (31)$$

is also the distribution of married couples at each step.

Consider first an agent whose current utility is below Λ . Eventually it meets an agent it likes by more than Λ , mutually, and marries. Its final utility is therefore p_0 conditioned on exceeding Λ , and the weight of this group is $F_0(\Lambda)$. An agent already at $x > \Lambda$ —necessarily unmarried, so its current partner is below *its* threshold—must instead clear x to move, giving p_0 conditioned on exceeding x . Integrating over x yields

$$p_\infty(u) = \Theta(u - \Lambda) p_0(u) \left[\int_\Lambda^u \frac{p_0(x)}{\bar{G}_0(x)} dx + \frac{F_0(\Lambda)}{\bar{G}_0(\Lambda)} \right]. \quad (32)$$

The remaining integral is elementary for *any* p_0 , since $p_0 = -d\bar{G}_0/dx$:

$$\int_\Lambda^u \frac{p_0(x)}{\bar{G}_0(x)} dx = \ln \frac{\bar{G}_0(\Lambda)}{\bar{G}_0(u)}. \quad (33)$$

So (32) is closed-form in general. It is normalised, since

$$\int_{-\infty}^{\infty} p_\infty(u) du = \bar{G}_0(\Lambda) + F_0(\Lambda) = 1. \quad (34)$$

For the uniform model, where $\bar{G}_0(x) = (2\sigma - x)/(4\sigma)$, this reduces to

$$p_\infty(u) = \Theta(u - \Lambda) p_0(u) \left[\ln \left(\frac{2\sigma - \Lambda}{2\sigma - u} \right) + \frac{\Lambda + 2\sigma}{2\sigma - \Lambda} \right]. \quad (35)$$

The first term of (35) diverges logarithmically as $u \rightarrow 2\sigma$, but integrably, so moments are well defined. For the uniform model the mean is

$$\mathbb{E}[U_\infty] = \frac{\Lambda^2}{16\sigma} + \frac{\Lambda}{4} + \frac{5\sigma}{4}, \quad (36)$$

which gives σ at $\Lambda = -2\sigma$, as it must—every agent marries the first partner it likes, and the average lands one sigma above the mean. It is increasing in Λ on the support, attaining 2σ at $\Lambda = 2\sigma$. Expression (36) is consistent with the scaling relation (8), since $\sigma \mathbb{E}[U_\infty] \big|_{\sigma=1, \Lambda} = \mathbb{E}[U_\infty] \big|_{\sigma, \sigma\Lambda}$.

Table 3 checks (36) against long simulations. The right panel of Figure 13 shows the same comparison graphically.

Two remarks. First, $\mathbb{E}[U_\infty]$ is increasing in Λ , which might naively suggest that a society should set its threshold as high as possible. That is not true at any finite horizon, as Figure 3 shows: at $t \approx 100$ the optimum is near $\Lambda = \sigma$, and $\Lambda = 1.5$ already does worse. The asymptotic regime is reached only by waiting arbitrarily long, which is a clean illustration of the Keynes remark quoted earlier. Second, the size of the prize is modest. At $\Lambda = \sigma$ the asymptotic value is 1.5625σ , and the dynamics reach 1.543σ in the uniform model by

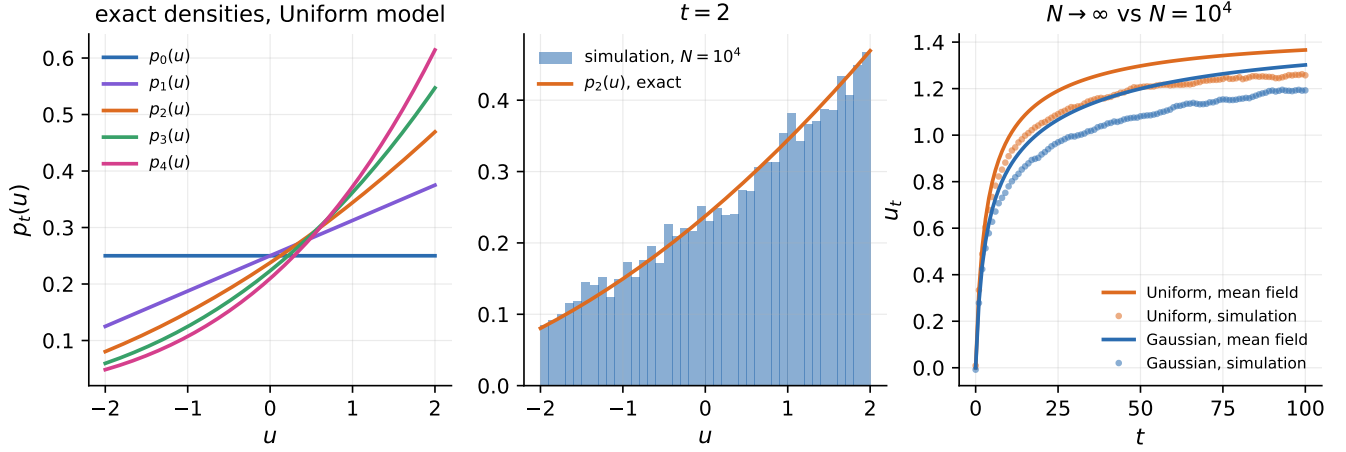


Figure 11: Left: exact densities $p_t(u)$ for $t = 0, \dots, 4$ in the uniform model. Centre: $p_2(u)$ against the simulated histogram at $N = 10,000$. Right: the mean-field mean utility against simulation for both models.

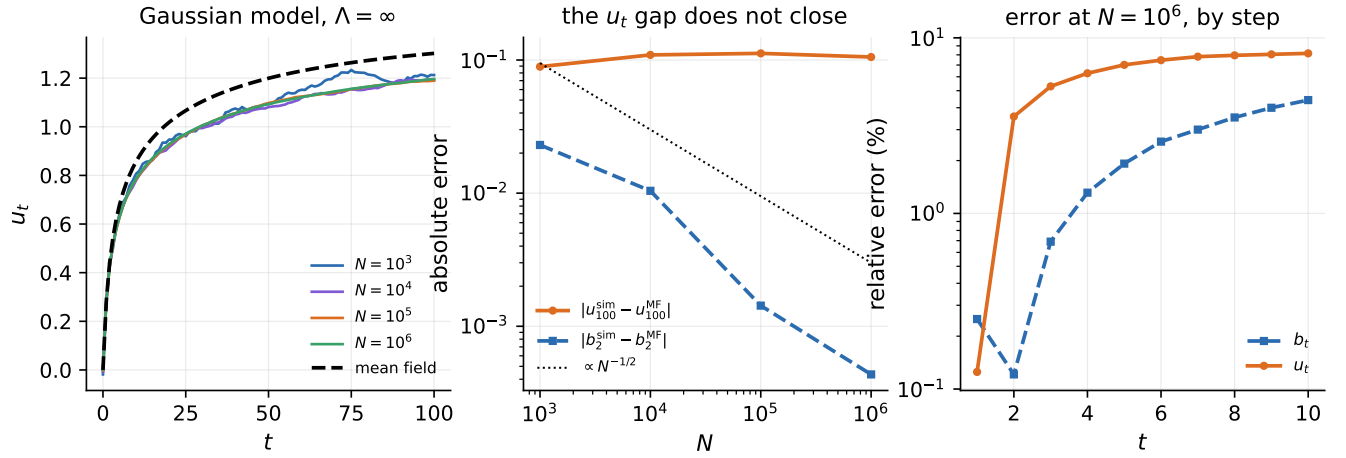


Figure 12: Left: average utility from simulation at increasing N , against the mean-field prediction. Centre: the error in u_{100} is flat in N while the error in b_2 falls as $N^{-1/2}$. Right: relative error at $N = 10^6$ by step; the deficit appears at $t = 2$.

Table 3: Asymptotic mean utility of married agents in the uniform model: Equation (36) against simulation at $N = 20,000$, $t = 800$, by which point 97% of agents are married.

Λ	Eq. (36)	Simulation
-2	1.0000	0.966
-1	1.0625	1.040
0	1.2500	1.240
1	1.5625	1.567
1.5	1.7656	1.770

$t = 100$ already. The long run is worth much less here than one might expect.

8 EMPIRICAL COMPARISON

Average utility is hard to measure in a real society; the number of married people is not. Comparing the share of married agents in the model with the share in existing societies therefore offers a way to test the model's adequacy. Figure 14 places the two side by side: the model's married share at each step for a range of Λ , and the share of men in England and Wales who had ever married by a given age, by birth cohort [32].

The agreement is arguably good, given how much the model deliberately ignores—divorce, most obviously. We note that divorce could be introduced by making affinities time-dependent, as discussed below. We also stress that the resemblance was not built into the matching process in any way: it is a purely dynamical consequence.

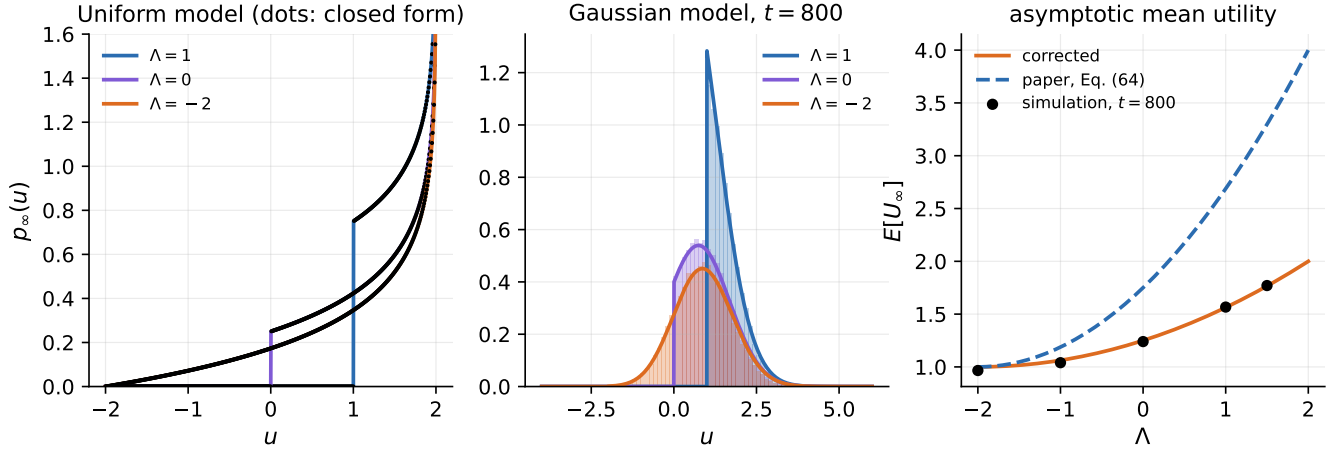


Figure 13: Left: $p_\infty(u)$ for the uniform model at $\sigma = 1$ and $\Lambda = 1, 0, -2$; dots mark the closed form (35). Centre: the same for the Gaussian model, against simulated histograms of married agents at $t = 800$. Right: the asymptotic mean against Λ .

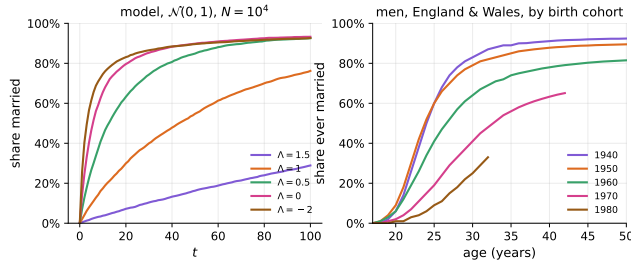


Figure 14: Left: share of married agents in the Gaussian model with $N = 10,000$ at each step, for $\Lambda = 1.5, 1, 0.5, 0, -2$ from the bottom line upwards. Right: share of men in England and Wales ever married by a given age, for the birth cohorts 1980, 1970, 1960, 1950, 1940 from the bottom line upwards [32].

Read through the model, the declining share of married men across cohorts corresponds to a society-wide shift to higher Λ . This agrees with rough expectations. In the past marriage was enforced by tradition and many couples married without high affinity; progressive movements weakened those forces and people became more demanding. If one goes further and associates each cohort with the value of Λ drawn in the same colour, then Figure 3 implies that average utility with respect to relationships rose from 1940 to 1970 and perhaps began to fall from 1970 to 1980. If approximately correct, this would suggest that progressive movements improved the matching market of traditional societies by raising Λ —but that Λ may since have gone too high and need rebalancing, in the direction the advice of Section 5 points.

Two caveats. First, steps are not years. The model's curve is concave from $t = 0$, whereas the data is S-shaped, because nobody marries before their late teens; if the initial steps are compressed, as is reasonable, the convexity of the early data is recovered. Any quantitative cohort-to- Λ mapping needs an explicit step-to-age map, which we do not have. Second, the association of cohorts with

particular Λ values in Figure 14 is a visual one, not a fit. Making it quantitative is the most promising route to the model's main empirical payoff: a way to access an unobservable property of a society, average utility with respect to relationships, through an observable one.

8.1 An Operational Rule for an Agent

To close, an operational answer to the question that opened the paper. The affinity matrix contains information that is not accessible to the agents: they discover an affinity only when they meet. An agent therefore does not know σ in advance. But it can use the utilities attained in previous matches to estimate a mean and a spread, and use those estimates to decide. Given utilities $\{u_t : 0 \leq t \leq T\}$, an agent should propose marriage at step $T + 1$ if

$$u_{T+1} > \frac{1}{T} \sum_{t=0}^T u_t + \sqrt{\frac{1}{T} \sum_{t=0}^T u_t^2 - \left(\frac{1}{T} \sum_{t=0}^T u_t\right)^2}. \quad (37)$$

It is left to the agent to estimate correctly its affinity with the agents it has already met. This is by no means a simple task, as we so commonly fool ourselves in matters of love; perhaps it is not even a well-defined quantity.

9 DISCUSSION AND LIMITATIONS

It is important to emphasise that the benefits of marriage disclosed by this model may be irrelevant once other aspects of marriage are considered. There are many layers of our preferences concerning relationships that the model does not represent, and Section 5.5 shows that even within the model the benefit is not uniform across agents.

A salient simplification is that affinities are time-independent. Making them time-dependent would bring divorce into the model for free: given a married couple, their affinity may fall below Λ , at which point they unmarry and may break apart if one of them finds another mate. Time-dependent affinities would also account for the spontaneous break-up of couples, which requires only that the

affinity fall below one partner's utility of being single. Understanding the average dynamical behaviour of affinity between agents is beyond the scope of this paper, but it is the single extension most likely to connect the model to further data, such as divorce rates.

Three further limitations are worth naming. The matching process is uniform over the whole society, whereas real meeting is strongly assortative by geography, education and social network. The threshold Λ is exogenous and uniform, whereas Section 5.5 suggests agents have heterogeneous incentives to choose it. And the mean-field theory of Section 7, as currently formulated, is exact only for the first step.

10 CONCLUSIONS

We proposed and studied an agent-based model of mate searching and marriage in large societies, built on a stochastic matching process and simple behavioural rules. Investigating it discloses some benefits of marriage as a search strategy. Most significantly, the average utility of a society with marriage can be higher than that of a society without it, with an optimum near $\Lambda \approx \sigma$; and marriage simultaneously compresses the distribution of outcomes. We summarised these results as what sounds like good advice: do not marry the first person you like and do not search for the love of your life, but get married if you like your partner more than a sigma above average.

We also showed that this benefit is not uniform. Once agents differ in desirability by more than sampling noise, the gain accrues to the less desirable, and beyond a moderate spread the most desirable agents are made strictly worse off. Marriage in this model is insurance against being left, and those never at risk of being left pay for it without needing it.

The average utility attained by the dynamics is well below that of a stable matching computed by Gale–Shapley, which can be taken as a simulation-based argument for a central mechanism—perhaps an application—with the information to coordinate the marriage market. To test the adequacy of the model for real societies we compared the evolution of the fraction of married agents with cohort data and obtained good agreement. Finally, we formulated the model in the limit of an infinite number of agents, derived the stationary distribution of married couples in closed form for arbitrary affinity distributions, and determined where the mean-field evolution equations are and are not valid.

This work leaves many directions open. For those with a mathematical bent, it would be valuable to derive the corrected large- N evolution equations by tracking the joint density of a couple, and to prove that they are the true limit. For those interested in the adequacy of the model, a more systematic comparison between the fraction of couples in the model and in real societies is needed, together with a mapping between steps and chronological time; with such a mapping one could infer Λ for a given society and, combined with Figure 3, advise it to be more or less selective. For those interested in the mechanism design question, Section 5.5 raises the issue of whether a society-level Λ is self-enforcing when agents differ in desirability. And for those interested in perfecting the model, the time-dependent affinity of Section 9 is the natural next step.

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